Table 1: Bridge compatibility rates (percent). Same settings, methods and previously selected scales as Table 1; no scale reselection. Means across three seeds.

Method	Weighted Discrete			Linear			Cubic			Smoothstep			2D Bridge		
	$Z_1 \uparrow$	$Z_2 \uparrow$	Both↑	$Z_1 \uparrow$	$Z_2 \uparrow$	Both↑	$Z_1 \uparrow$	$Z_2 \uparrow$	Both↑	$Z_1 \uparrow$	$Z_2 \uparrow$	Both↑	$Z_1 \uparrow$	$Z_2 \uparrow$	Both↑
Conditional diffusion															
CD	23.25	95.35	18.60	91.99	91.29	83.28	93.38	83.94	77.31	91.55	94.08	85.63	88.93	88.12	82.36
Compositional CD	86.40	95.60	82.00	71.21	71.24	42.45	63.40	69.71	33.11	56.02	80.05	36.07	46.42	52.64	34.65
Classifier-guided diffusion															
CG: sparse factorized	90.20	92.43	82.63	74.87	71.48	46.35	77.29	66.57	43.88	64.47	87.63	52.10	60.07	70.80	47.28
CG: dense factorized	97.08	90.05	**87.13**	75.60	72.15	47.75	75.47	66.42	41.89	62.03	86.83	48.86	67.50	62.78	47.33
CG: joint	86.07	31.63	17.78	76.38	83.14	59.52	70.08	81.53	51.63	71.32	91.99	63.31	71.21	79.28	59.93
Extrapolation-inspired baselines															
Additive mean†	0.00	**100.00**	0.00	**100.00**	**100.00**	**100.00**	**100.00**	0.00	0.00	0.00	**100.00**	0.00	0.00	0.00	0.00
Additive + residual†	49.15	52.40	10.33	55.05	54.07	9.23	64.16	43.77	9.36	49.17	60.92	10.35	21.57	22.98	2.08
Fixed composition + residual†	49.88	52.33	9.87	36.70	70.46	9.41	54.10	55.94	10.29	32.80	74.38	9.54	17.48	26.45	1.99
Rep4Ex-inspired + residual†	39.53	66.00	9.30	53.99	40.79	7.18	62.08	32.57	5.79	51.42	50.28	11.98	24.53	28.61	3.37
SEDGE (ours)															
SEDGE: no weighting	**97.35**	14.50	13.13	98.73	9.70	9.02	98.70	9.73	9.05	**98.73**	9.65	8.97	**97.41**	2.62	2.46
SEDGE-LGD	78.95	81.93	60.88	89.27	88.36	77.65	73.73	98.16	71.89	88.66	90.53	79.18	84.68	84.96	77.85
SEDGE-LWS	89.35	93.82	83.17	92.58	**100.00**	92.58	93.00	**100.00**	**93.00**	92.15	**100.00**	**92.15**	89.47	**94.03**	**89.21**

Compatibility asks whether some legal private variable can satisfy the requested specification at the generated bridge, not whether the generated private variable does so. Both requires the same bridge to permit both specifications. Out-of-support or nonfinite bridges count as incompatible. For continuous mechanisms $Z_i = a_i(C)X_i + b_i(C)$, $e_i = \max\{b_i(C) - z_i, z_i - b_i(C) - a_i(C), 0\}$. Binary e_i is the incompatibility indicator. Residual means condition on bridge root support; support rates and actual full-sample validity are in the companion CSV. Residuals use raw specification units; compare within each DGP. Bold indicates the best displayed mean, not significance. †Paper-inspired adaptations.

Table 2: Bridge compatibility rates (percent). Same settings, methods and previously selected scales as Table 1; no scale reselection. Mean and sample standard deviation across three seeds.

Method	Weighted Discrete			Linear			Cubic			Smoothstep			2D Bridge		
	$Z_1 \uparrow$	$Z_2 \uparrow$	Both↑	$Z_1 \uparrow$	$Z_2 \uparrow$	Both↑	$Z_1 \uparrow$	$Z_2 \uparrow$	Both↑	$Z_1 \uparrow$	$Z_2 \uparrow$	Both↑	$Z_1 \uparrow$	$Z_2 \uparrow$	Both↑
Conditional diffusion															
CD	23.25 ± 0.18	95.35 ± 1.80	18.60 ± 1.63	91.99 ± 1.01	91.29 ± 2.85	83.28 ± 3.29	93.38 ± 3.05	83.94 ± 4.69	77.31 ± 7.70	91.55 ± 1.38	94.08 ± 2.64	85.63 ± 3.79	88.93 ± 5.02	88.12 ± 5.57	82.36 ± 8.36
Compositional CD	86.40 ± 5.51	95.60 ± 2.76	82.00 ± 2.83	71.21 ± 2.00	71.24 ± 1.98	42.45 ± 1.12	63.40 ± 3.48	69.71 ± 4.56	33.11 ± 1.13	56.02 ± 1.62	80.05 ± 1.56	36.07 ± 0.15	46.42 ± 2.04	52.64 ± 6.27	34.65 ± 2.17
Classifier-guided diffusion															
CG: sparse factorized	90.20 ± 5.80	92.43 ± 3.03	82.63 ± 6.16	74.87 ± 6.50	71.48 ± 7.46	46.35 ± 1.33	77.29 ± 1.68	66.57 ± 1.39	43.88 ± 2.93	64.47 ± 3.40	87.63 ± 2.31	52.10 ± 1.74	60.07 ± 6.39	70.80 ± 4.57	47.28 ± 1.93
CG: dense factorized	97.08 ± 1.07	90.05 ± 4.29	**87.13** ± 5.14	75.60 ± 1.23	72.15 ± 1.80	47.75 ± 1.81	75.47 ± 2.64	66.42 ± 3.06	41.89 ± 1.18	62.03 ± 5.31	86.83 ± 9.16	48.86 ± 4.32	67.50 ± 2.19	62.78 ± 8.45	47.33 ± 5.30
CG: joint	86.07 ± 6.83	31.63 ± 12.01	17.78 ± 5.31	76.38 ± 8.84	83.14 ± 5.91	59.52 ± 2.96	70.08 ± 3.82	81.53 ± 4.58	51.63 ± 2.67	71.32 ± 1.71	91.99 ± 1.97	63.31 ± 1.13	71.21 ± 2.81	79.28 ± 8.22	59.93 ± 7.60
Extrapolation-inspired baselines															
Additive mean†	0.00 ± 0.00	**100.00** ± 0.00	0.00 ± 0.00	**100.00** ± 0.00	**100.00** ± 0.00	**100.00** ± 0.00	**100.00** ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	**100.00** ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00
Additive + residual†	49.15 ± 0.58	52.40 ± 1.21	10.33 ± 0.46	55.05 ± 1.25	54.07 ± 0.83	9.23 ± 0.54	64.16 ± 0.87	43.77 ± 1.28	9.36 ± 0.17	49.17 ± 1.05	60.92 ± 1.33	10.35 ± 0.81	21.57 ± 1.05	22.98 ± 0.46	2.08 ± 0.44
Fixed composition + residual†	49.88 ± 0.38	52.33 ± 0.96	9.87 ± 0.35	36.70 ± 1.34	70.46 ± 1.08	9.41 ± 0.66	54.10 ± 1.52	55.94 ± 1.96	10.29 ± 0.66	32.80 ± 2.99	74.38 ± 2.61	9.54 ± 0.66	17.48 ± 3.83	26.45 ± 2.32	1.99 ± 0.48
Rep4Ex-inspired + residual†	39.53 ± 24.39	66.00 ± 25.55	9.30 ± 1.71	53.99 ± 23.97	40.79 ± 28.93	7.18 ± 5.13	62.08 ± 8.99	32.57 ± 16.39	5.79 ± 2.82	51.42 ± 23.05	50.28 ± 27.71	11.98 ± 8.09	24.53 ± 19.76	28.61 ± 12.30	3.37 ± 4.38
SEDGE (ours)															
SEDGE: no weighting	**97.35** ± 0.23	14.50 ± 0.22	13.13 ± 0.13	98.73 ± 0.48	9.70 ± 1.32	9.02 ± 1.49	98.70 ± 0.52	9.73 ± 1.36	9.05 ± 1.51	**98.73** ± 0.48	9.65 ± 1.34	8.97 ± 1.51	**97.41** ± 0.35	2.62 ± 0.53	2.46 ± 0.49
SEDGE-LGD	78.95 ± 8.34	81.93 ± 6.32	60.88 ± 4.38	89.27 ± 2.61	88.36 ± 4.28	77.65 ± 2.62	73.73 ± 4.65	98.16 ± 0.90	71.89 ± 5.39	88.66 ± 2.89	90.53 ± 5.65	79.18 ± 4.65	84.68 ± 2.08	84.96 ± 1.80	77.85 ± 2.61
SEDGE-LWS	89.35 ± 2.15	93.82 ± 3.48	83.17 ± 2.13	92.58 ± 1.31	**100.00** ± 0.00	92.58 ± 1.31	93.00 ± 1.15	**100.00** ± 0.00	**93.00** ± 1.15	92.15 ± 1.10	**100.00** ± 0.00	**92.15** ± 1.10	89.47 ± 1.86	**94.03** ± 1.65	**89.21** ± 1.49

Compatibility asks whether some legal private variable can satisfy the requested specification at the generated bridge, not whether the generated private variable does so. Both requires the same bridge to permit both specifications. Out-of-support or nonfinite bridges count as incompatible. For continuous mechanisms $Z_i = a_i(C)X_i + b_i(C)$, $e_i = \max\{b_i(C) - z_i, z_i - b_i(C) - a_i(C), 0\}$. Binary e_i is the incompatibility indicator. Residual means condition on bridge root support; support rates and actual full-sample validity are in the companion CSV. Residuals use raw specification units; compare within each DGP. Bold indicates the best displayed mean, not significance. †Paper-inspired adaptations.

Table 3: Mean irreducible specification residuals at generated bridge coordinates. Same settings, methods and previously selected scales as Table 1; no scale reselection. Means across three seeds.

Method	Weighted Discrete			Linear			Cubic			Smoothstep			2D Bridge		
	$e_1 \downarrow$	$e_2 \downarrow$	$\max(e_1, e_2) \downarrow$	$e_1 \downarrow$	$e_2 \downarrow$	$\max(e_1, e_2) \downarrow$	$e_1 \downarrow$	$e_2 \downarrow$	$\max(e_1, e_2) \downarrow$	$e_1 \downarrow$	$e_2 \downarrow$	$\max(e_1, e_2) \downarrow$	$e_1 \downarrow$	$e_2 \downarrow$	$\max(e_1, e_2) \downarrow$
Conditional diffusion															
CD	0.7675	0.0465	0.8140	0.0003	0.0003	0.0006	0.0002	0.0008	0.0010	0.0003	0.0002	0.0005	0.0003	0.0004	0.0007
Compositional CD	0.1360	0.0440	0.1800	0.0034	0.0033	0.0067	0.0092	0.0033	0.0124	0.0077	0.0029	0.0106	0.0105	0.0093	0.0129
Classifier-guided diffusion															
CG: sparse factorized	0.0980	0.0757	0.1737	0.0036	0.0045	0.0081	0.0042	0.0046	0.0088	0.0064	0.0014	0.0078	0.0083	0.0051	0.0110
CG: dense factorized	0.0292	0.0995	**0.1287**	0.0035	0.0040	0.0075	0.0046	0.0045	0.0091	0.0069	0.0019	0.0088	0.0055	0.0077	0.0108
CG: joint	0.1386	0.6834	0.8220	0.0026	0.0017	0.0042	0.0040	0.0020	0.0060	0.0034	0.0007	0.0041	0.0042	0.0042	0.0076
Extrapolation-inspired baselines															
Additive mean†	1.0000	**0.0000**	1.0000	**0.0000**	**0.0000**	**0.0000**	**0.0000**	0.0144	0.0144	0.0045	**0.0000**	0.0045	0.0424	0.0446	0.0476
Additive + residual†	0.4611	0.4256	0.8867	0.0402	0.0403	0.0806	0.0273	0.0224	0.0497	0.0470	0.0405	0.0875	0.0939	0.0939	0.1621
Fixed composition + residual†	0.4598	0.4333	0.8931	0.0674	0.0177	0.0851	0.0437	0.0160	0.0597	0.0709	0.0193	0.0902	0.1098	0.0764	0.1615
Rep4Ex-inspired + residual†	0.5840	0.3184	0.9025	0.0346	0.0482	0.0828	0.0296	0.0284	0.0580	0.0373	0.0523	0.0896	0.1016	0.0640	0.1466
SEDGE (ours)															
SEDGE: no weighting	**0.0138**	0.8531	0.8670	0.0000	0.1287	0.1287	0.0000	0.0437	0.0437	**0.0000**	0.1578	0.1579	**0.0000**	0.3458	0.3458
SEDGE-LGD	0.2105	0.1807	0.3912	0.0004	0.0044	0.0048	0.0008	0.0007	0.0015	0.0004	0.0052	0.0056	0.0003	0.0288	0.0292
SEDGE-LWS	0.1065	0.0618	0.1683	0.0002	**0.0000**	0.0002	0.0002	**0.0000**	**0.0002**	0.0003	**0.0000**	**0.0003**	0.0002	**0.0000**	**0.0002**

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Table 4: Mean irreducible specification residuals at generated bridge coordinates. Same settings, methods and previously selected scales as Table 1; no scale reselection. Mean and sample standard deviation across three seeds.

Method	Weighted Discrete			Linear			Cubic			Smoothstep			2D Bridge		
	$e_1 \downarrow$	$e_2 \downarrow$	$\max(e_1, e_2) \downarrow$	$e_1 \downarrow$	$e_2 \downarrow$	$\max(e_1, e_2) \downarrow$	$e_1 \downarrow$	$e_2 \downarrow$	$\max(e_1, e_2) \downarrow$	$e_1 \downarrow$	$e_2 \downarrow$	$\max(e_1, e_2) \downarrow$	$e_1 \downarrow$	$e_2 \downarrow$	$\max(e_1, e_2) \downarrow$
Conditional diffusion															
CD	0.7675 ± 0.0018	0.0465 ± 0.0180	0.8140 ± 0.0163	0.0003 ± 0.0000	0.0003 ± 0.0001	0.0006 ± 0.0002	0.0002 ± 0.0001	0.0008 ± 0.0003	0.0010 ± 0.0004	0.0003 ± 0.0001	0.0002 ± 0.0001	0.0005 ± 0.0002	0.0003 ± 0.0003	0.0004 ± 0.0003	0.0007 ± 0.0005
Compositional CD	0.1360 ± 0.0551	0.0440 ± 0.0276	0.1800 ± 0.0283	0.0034 ± 0.0004	0.0033 ± 0.0006	0.0067 ± 0.0004	0.0092 ± 0.0009	0.0033 ± 0.0008	0.0124 ± 0.0005	0.0077 ± 0.0004	0.0029 ± 0.0005	0.0106 ± 0.0001	0.0105 ± 0.0015	0.0093 ± 0.0025	0.0129 ± 0.0002
Classifier-guided diffusion															
CG: sparse factorized	0.0980 ± 0.0580	0.0757 ± 0.0303	0.1737 ± 0.0616	0.0036 ± 0.0013	0.0045 ± 0.0015	0.0081 ± 0.0003	0.0042 ± 0.0004	0.0046 ± 0.0008	0.0088 ± 0.0010	0.0064 ± 0.0010	0.0014 ± 0.0003	0.0078 ± 0.0007	0.0083 ± 0.0031	0.0051 ± 0.0013	0.0110 ± 0.0023
CG: dense factorized	0.0292 ± 0.0107	0.0995 ± 0.0429	**0.1287** ± 0.0514	0.0035 ± 0.0006	0.0040 ± 0.0002	0.0075 ± 0.0006	0.0046 ± 0.0003	0.0045 ± 0.0006	0.0091 ± 0.0005	0.0069 ± 0.0008	0.0019 ± 0.0018	0.0088 ± 0.0014	0.0055 ± 0.0006	0.0077 ± 0.0029	0.0108 ± 0.0025
CG: joint	0.1386 ± 0.0687	0.6834 ± 0.1201	0.8220 ± 0.0531	0.0026 ± 0.0013	0.0017 ± 0.0007	0.0042 ± 0.0006	0.0040 ± 0.0005	0.0020 ± 0.0008	0.0060 ± 0.0003	0.0034 ± 0.0003	0.0007 ± 0.0001	0.0041 ± 0.0002	0.0042 ± 0.0005	0.0042 ± 0.0003	0.0076 ± 0.0007
Extrapolation-inspired baselines															
Additive mean†	1.0000 ± 0.0000	**0.0000** ± 0.0000	1.0000 ± 0.0000	**0.0000** ± 0.0000	**0.0000** ± 0.0000	**0.0000** ± 0.0000	**0.0000** ± 0.0000	0.0144 ± 0.0031	0.0144 ± 0.0031	0.0045 ± 0.0038	**0.0000** ± 0.0000	0.0045 ± 0.0038	0.0424 ± 0.0029	0.0446 ± 0.0074	0.0476 ± 0.0021
Additive + residual†	0.4611 ± 0.0087	0.4256 ± 0.0104	0.8867 ± 0.0059	0.0402 ± 0.0022	0.0403 ± 0.0017	0.0806 ± 0.0024	0.0273 ± 0.0009	0.0224 ± 0.0008	0.0497 ± 0.0004	0.0470 ± 0.0031	0.0405 ± 0.0020	0.0875 ± 0.0030	0.0939 ± 0.0018	0.0939 ± 0.0020	0.1621 ± 0.0003
Fixed composition + residual†	0.4598 ± 0.0095	0.4333 ± 0.0046	0.8931 ± 0.0049	0.0674 ± 0.0016	0.0177 ± 0.0014	0.0851 ± 0.0013	0.0437 ± 0.0018	0.0160 ± 0.0012	0.0597 ± 0.0010	0.0709 ± 0.0041	0.0193 ± 0.0030	0.0902 ± 0.0020	0.1098 ± 0.0106	0.0764 ± 0.0179	0.1615 ± 0.0054
Rep4Ex-inspired + residual†	0.5840 ± 0.2577	0.3184 ± 0.2385	0.9025 ± 0.0232	0.0346 ± 0.0221	0.0482 ± 0.0326	0.0828 ± 0.0309	0.0296 ± 0.0064	0.0284 ± 0.0097	0.0580 ± 0.0040	0.0373 ± 0.0230	0.0523 ± 0.0558	0.0896 ± 0.0436	0.1016 ± 0.1032	0.0640 ± 0.0208	0.1466 ± 0.0768
SEDGE (ours)															
SEDGE: no weighting	**0.0138** ± 0.0013	0.8531 ± 0.0024	0.8670 ± 0.0014	0.0000 ± 0.0000	0.1287 ± 0.0010	0.1287 ± 0.0011	0.0000 ± 0.0000	0.0437 ± 0.0004	0.0437 ± 0.0004	**0.0000** ± 0.0000	0.1578 ± 0.0012	0.1579 ± 0.0013	**0.0000** ± 0.0000	0.3458 ± 0.0019	0.3458 ± 0.0019
SEDGE-LGD	0.2105 ± 0.0834	0.1807 ± 0.0632	0.3912 ± 0.0438	0.0004 ± 0.0000	0.0044 ± 0.0004	0.0048 ± 0.0004	0.0008 ± 0.0002	0.0007 ± 0.0004	0.0015 ± 0.0005	0.0004 ± 0.0000	0.0052 ± 0.0014	0.0056 ± 0.0014	0.0003 ± 0.0001	0.0288 ± 0.0163	0.0292 ± 0.0163
SEDGE-LWS	0.1065 ± 0.0215	0.0618 ± 0.0348	0.1683 ± 0.0213	0.0002 ± 0.0001	**0.0000** ± 0.0000	0.0002 ± 0.0001	0.0002 ± 0.0000	**0.0000** ± 0.0000	**0.0002** ± 0.0000	0.0003 ± 0.0001	**0.0000** ± 0.0000	**0.0003** ± 0.0001	0.0002 ± 0.0001	**0.0000** ± 0.0000	**0.0002** ± 0.0001

Compatibility asks whether some legal private variable can satisfy the requested specification at the generated bridge, not whether the generated private variable does so. Both requires the same bridge to permit both specifications. Out-of-support or nonfinite bridges count as incompatible. For continuous mechanisms $Z_i = a_i(C)X_i + b_i(C)$, $e_i = \max\{b_i(C) - z_i, z_i - b_i(C) - a_i(C), 0\}$. Binary e_i is the incompatibility indicator. Residual means condition on bridge root support; support rates and actual full-sample validity are in the companion CSV. Residuals use raw specification units; compare within each DGP. Bold indicates the best displayed mean, not significance. †Paper-inspired adaptations.